

YELETS, Russian Federation

WMO: 279280

Lat: 52.633N

Lon: 38.517E

Elev: 167

StdP: 99.33

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.2	-21.2	-27.0	0.3	-24.1	-24.0	0.4	-20.8	6.9	-4.2	6.3	-4.9	1.4	30	0.501

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	31.6	19.6	29.7	19.1	27.9	18.4	21.2	28.1	20.4	26.9	19.6	25.8	2.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	14.1	23.6	18.2	13.4	22.8	17.5	12.8	22.2	62.2	28.2	59.2	27.1	56.6	26.0	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
6.2	5.4	4.9		DB	-26.8	33.7	3.3	2.2	-29.2	35.3	-31.1	36.6	-32.9	37.9	-35.3	39.5
				WB	-27.0	22.5	3.2	1.0	-29.3	23.2	-31.2	23.8	-33.0	24.3	-35.4	25.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.8	-7.1	-6.8	-1.3	8.0	15.0	18.3	20.7	19.2	13.3	6.7	0.0	-4.8	(6)
(7)		DBStd	11.26	6.80	6.84	4.93	4.73	4.72	3.79	3.41	3.80	4.23	4.48	5.31	6.32	(7)
(8)		HDD10.0	2355	531	469	352	95	11	1	0	0	16	121	300	459	(8)
(9)		HDD18.3	4421	789	703	610	311	126	48	14	35	157	360	550	718	(9)
(10)		CDD10.0	1201	0	0	0	33	165	249	333	284	116	19	1	0	(10)
(11)		CDD18.3	224	0	0	0	0	21	46	89	61	7	0	0	0	(11)
(12)		CDH23.3	2137	0	0	0	6	221	407	793	626	85	0	0	0	(12)
(13)		CDH26.7	682	0	0	0	1	52	113	276	228	13	0	0	0	(13)
(14)	Wind	WSAvg	2.3	2.6	2.6	2.6	2.5	2.2	2.0	1.8	1.8	1.9	2.3	2.5	2.7	(14)
(15)	Precipitation	PrecAvg	578	42	36	33	40	44	66	73	57	52	54	43	39	(15)
(16)		PrecMax	827	101	75	63	81	106	143	125	186	129	139	108	112	(16)
(17)		PrecMin	344	16	8	8	3	3	10	7	8	3	8	3	9	(17)
(18)		PrecStd	103	18	16	17	22	24	33	30	42	33	31	26	23	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.2	3.8	14.0	24.3	30.5	32.2	34.7	35.7	28.5	20.9	12.7	7.7	(19)
(20)		MCWB	3.2	2.3	7.8	13.0	18.1	18.7	20.1	19.5	16.2	13.8	10.9	6.6		(20)
(21)		2%	DB	2.5	2.7	9.3	21.0	28.2	29.7	31.6	31.5	26.0	17.8	10.2	5.3	(21)
(22)		MCWB	1.6	1.6	5.2	12.1	16.9	18.9	20.2	19.6	15.8	13.0	9.0	4.5		(22)
(23)		5%	DB	1.6	1.9	6.7	18.7	25.9	27.6	29.8	29.1	23.4	15.1	8.4	3.6	(23)
(24)		MCWB	0.9	1.0	3.6	11.0	16.3	18.6	19.9	18.9	15.2	11.7	7.2	2.9		(24)
(25)		10%	DB	0.9	1.2	4.6	16.3	23.5	25.6	27.7	26.7	20.7	13.1	6.6	1.9	(25)
(26)		MCWB	0.3	0.5	2.6	9.9	15.3	17.7	19.1	17.8	14.5	10.6	5.6	1.2		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.4	2.9	8.2	14.4	19.6	21.5	22.4	22.1	18.2	15.4	11.0	7.0	(27)
(28)		MCDWB	4.0	3.5	12.5	20.1	27.4	27.2	30.1	29.2	24.0	18.8	12.5	7.7		(28)
(29)		2%	WB	1.9	1.9	5.9	12.9	18.2	20.4	21.4	20.8	17.1	13.8	9.2	4.6	(29)
(30)		MCDWB	2.3	2.5	8.7	18.8	25.7	26.5	28.6	27.9	22.9	16.6	10.1	5.2		(30)
(31)		5%	WB	1.0	1.2	4.2	11.7	17.1	19.5	20.6	19.7	16.0	12.2	7.3	3.0	(31)
(32)		MCDWB	1.5	1.6	6.2	17.4	23.6	25.4	27.3	26.9	20.9	14.8	8.2	3.5		(32)
(33)		10%	WB	0.3	0.6	2.7	10.5	16.0	18.7	19.9	18.8	14.9	10.8	5.7	1.4	(33)
(34)		MCDWB	0.9	1.2	4.3	15.5	22.2	24.1	26.0	25.2	19.9	12.8	6.5	1.8		(34)
(35)	Mean Daily Temperature Range	MDBR	5.1	5.9	6.6	9.6	10.7	9.9	10.2	11.2	9.6	6.9	4.5	4.6		(35)
(36)		5% DB	MCDBR	3.9	4.1	9.3	13.3	13.6	13.0	13.1	14.6	14.3	10.8	6.4	4.0	(36)
(37)		MCWBR	3.7	3.7	5.8	6.3	5.3	4.9	4.3	5.1	6.0	6.3	5.4	3.7		(37)
(38)		5% WB	MCDBR	3.7	3.9	8.4	11.7	12.1	10.9	11.3	12.5	11.4	9.1	5.7	3.8	(38)
(39)		MCWBR	3.8	3.8	5.6	6.1	5.3	4.6	4.4	4.9	5.7	6.0	5.1	3.6		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.298	0.335	0.359	0.435	0.395	0.386	0.420	0.438	0.388	0.360	0.334	0.302		(40)
(41)		taud	2.117	2.027	2.084	2.084	2.308	2.348	2.268	2.203	2.321	2.342	2.258	2.152		(41)
(42)		Ebn at Noon	658	738	812	793	856	866	826	784	784	724	620	579		(42)
(43)		Edh at Noon	73	108	126	144	121	118	125	127	100	80	65	60		(43)
(44)	All-Sky Solar Radiation	RadAvg	0.89	1.90	3.36	4.20	5.69	5.87	5.76	4.90	3.23	1.65	0.70	0.49		(44)
(45)		RadStd	0.21	0.22	0.53	0.36	0.45	0.58	0.52	0.46	0.41	0.24	0.14	0.15		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.86	N/A	N/A	N/A	N/A	N/A	-162	-254	+136	N/A
(47)	Regional (9 neighbors)	+0.80	N/A	N/A	N/A	N/A	N/A	-141	-208	+118	+42

Nomenclature: See separate page